Department of Computer Science & Engineering

PES University, Bengaluru

Software Engineering Lab (SE Lab) — Lab 2 Report

Agile Backlog Creation & Sprint Simulation in Jira
Student Information:

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- Problem Statement: #48 — Incident Escalation & On-Call Rotation Engine
- GitHub Repository: github.com/arjunsrikanthh/PES1UG24AM920_se_lab
- Jira Workspace: arjun-srikanth.atlassian.net (Project: Software-development / SD board)

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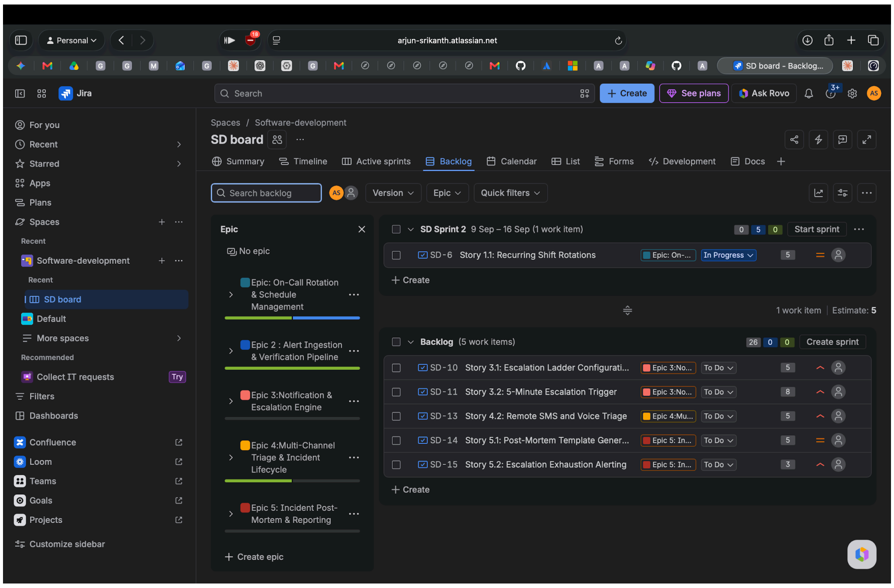
1. Backlog, Epics & Story Point Assignments

Functional requirements from Lab 1 were decomposed into five major Epics and prioritized User Stories in Jira. Effort estimation was conducted using Planning Poker based on the Fibonacci scale (1, 2, 3, 5, 8):

- Epic 1: On-Call Rotation & Schedule Management
  - SD-6: Story 1.1: Recurring Shift Rotations (5 Story Points | Medium Priority)
  - SD-7: Story 1.2: Shift Overrides & Holiday Swaps (3 Story Points | Medium Priority)
- Epic 2: Alert Ingestion & Verification Pipeline
  - SD-8: Story 2.1: Webhook Alert Ingestion (5 Story Points | High Priority)
  - SD-9: Story 2.2: Active Responder Resolution (3 Story Points | High Priority)
- Epic 3: Notification & Escalation Engine
  - SD-10: Story 3.1: Escalation Ladder Configuration (5 Story Points | High Priority)
  - SD-11: Story 3.2: 5-Minute Escalation Trigger (8 Story Points | High Priority)
- Epic 4: Multi-Channel Triage & Incident Lifecycle

    - SD-12: Story 4.1: Web Dashboard Triage (5 Story Points | High Priority)
    - SD-13: Story 4.2: Remote SMS and Voice Triage (5 Story Points | High Priority)
- Epic 5: Incident Post-Mortem & Reporting
    - SD-14: Story 5.1: Post-Mortem Template Generation (5 Story Points | Medium Priority)
    - SD-15: Story 5.2: Escalation Exhaustion Alerting (3 Story Points | Medium Priority)



Deliverable Screenshot 1: Backlog with Epics & Story Point Assignments

Figure 1: Backlog view in Jira showing Epics, User Stories, and Story Point estimates.

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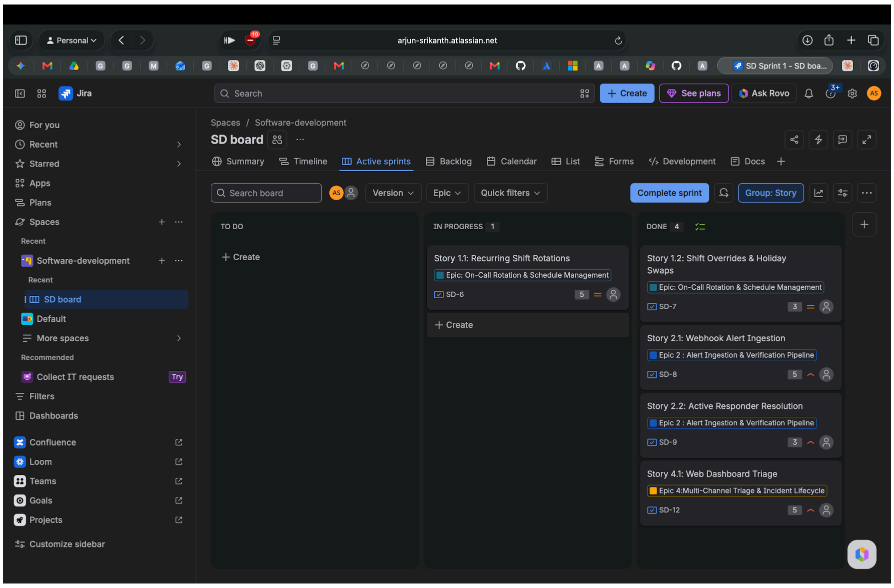
2. Sprint Board (Active Sprint View)

For the sprint simulation, SD Sprint 1 was created with a 1-week timebox. Five user stories totaling 21 story points were committed (SD-6, SD-7, SD-8, SD-9, SD-12). Work items were progressed across To Do, In Progress, and Done:

- Completed Items (16 Story Points):
    - SD-7: Shift Overrides & Holiday Swaps (3 pts) — Moved to Done
    - SD-8: Webhook Alert Ingestion (5 pts) — Moved to Done
    - SD-9: Active Responder Resolution (3 pts) — Moved to Done

    - SD-12: Web Dashboard Triage (5 pts) — Moved to Done
- In-Progress Item (5 Story Points):
    - SD-6: Recurring Shift Rotations (5 pts) — In Progress at sprint closure; carried over to SD Sprint 2 (9 Sep – 16 Sep).



Deliverable Screenshot 2: Active Sprint Board

Figure 2: Active Sprint board for SD Sprint 1 showing In Progress and Done work items.

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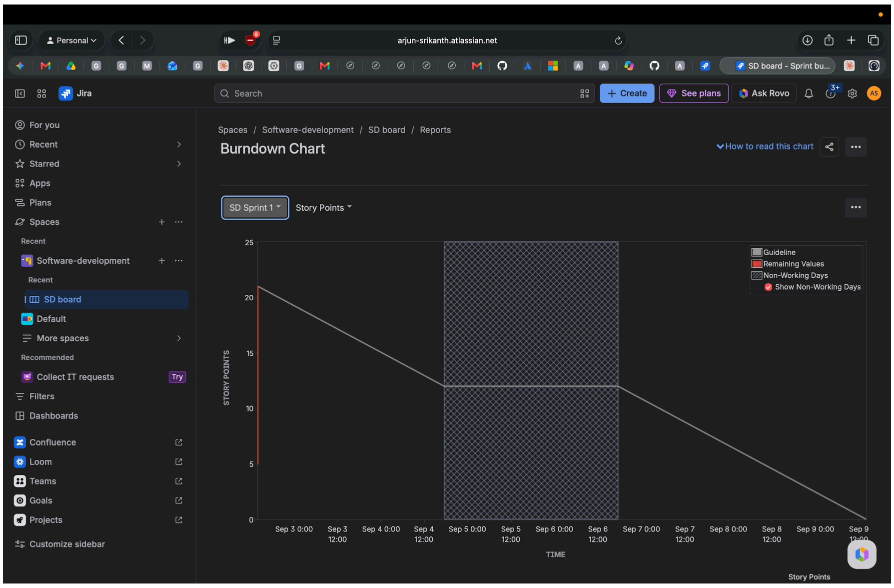
3. Burndown Chart & Sprint Progress

The burndown chart tracks story point completion relative to the linear guideline:

- Initial commitment: 21 story points at sprint start (2:02 PM).
- Burned down: 16 points completed across SD-7, SD-8, SD-9, and SD-12 (2:03 PM).
- Remaining: 5 points (SD-6) at sprint closure (2:05 PM).
- Delivered velocity for SD Sprint 1: 16 story points.



Deliverable Screenshot 3: Burndown Chart

Figure 3: Burndown Chart for SD Sprint 1 showing story points completed versus the guideline.

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4. Answers to Reflection Questions

Question 1: Did your estimations reflect the actual effort?

The Fibonacci estimates accurately reflected the relative differences in complexity across most stories. Straightforward tasks like responder lookup (SD-9, 3 points) and shift overrides (SD-7, 3 points) were completed smoothly. However, SD-6 (Recurring Shift Rotations, estimated at 5 points) took longer than expected due to complex scheduling math and edge cases, remaining in progress at sprint completion. This indicates that recurring schedule logic carried higher uncertainty and should have been sized at 8 points or split into smaller stories.

Question 2: Was your backlog well-prioritized?

Yes. The backlog was prioritized strictly by functional dependency and operational risk. Ingestion (Epic 2) and rotation scheduling (Epic 1) formed the foundation required for alert dispatch. Web triage (SD-12) was included to provide immediate end-to-end visibility. Dependent features such as automated multi-tier escalation triggers (Epic 3), remote voice/SMS triage (Epic 4), and post-mortem reporting (Epic 5) were properly deferred to future sprints.

Question 3: How did your simulated sprint align with your plan?

The team planned 21 story points and delivered 16 story points (4 out of 5 stories completed, representing a 76.2% completion rate). The simulation demonstrated realistic Scrum dynamics: rather than rushing or partially accepting an incomplete feature, SD-6 was cleanly kept in progress and rolled over into Sprint 2.

Question 4: What insights did the burndown chart give about your team's capacity?

The burndown chart showed that the team's sustainable velocity for a 1-week sprint is approximately 16 story points. Committing 21 points exceeded team capacity. For future sprint planning (such as SD Sprint 2), 15–16 story points will serve as a grounded capacity baseline. Furthermore, the sharp drops in the burndown curve reflect batch completion during the simulation, reinforcing the need for continuous daily task updates in actual engineering workflows.